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PARTICLE SEPARATION SYSTEM UPSTREAM OF A HEAT EXCHANGER IN A GAS TURBINE ENGINE

Abstract

A particle separation system includes at least one inertial particle separator spaced apart from or integrated as a unitary piece with a heat exchanger. The at least one inertial particle separator is disposed upstream of the heat exchanger. The at least one inertial particle separator is configured and arranged to direct a first fluid flow containing a first amount of particles to an inner passage provided at a hub of the heat exchanger, an outer passage provided at a tip of the heat exchanger, or both, the hub and the tip being located at a radial periphery of the heat exchanger, and to direct a second fluid flow containing a second amount of particles to a central portion of the heat exchanger to cool down a fluid circulating within the heat exchanger. The second amount of particles is substantially less than the first amount of particles.

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Background/Summary

FIELD

[0001] The present disclosure relates to a particle separation system disposed upstream of a heat exchanger in a gas turbine engine.

BACKGROUND

[0002] A gas turbine engine generally includes a turbo-engine and a rotor assembly. Gas turbine engines, such as turbofan engines, may be used for aircraft propulsion. In the case of a turbofan engine, the rotor assembly may be configured as a fan assembly. Gas turbine engines generally include a thermal management system to manage thermal loads during operation of the gas turbine engine. The thermal management system includes a heat exchanger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** is a schematic cross-sectional view of a gas turbine engine, according to embodiment of the present disclosure.

[0004] FIG. **2**A shows a location of a heat exchanger and a particle separation system within a fan duct of the gas turbine engine, according to an embodiment of the present disclosure.

[0005] FIG. **2**B shows a location of a heat exchanger along with a particle separation system within the fan duct of the gas turbine engine, according to another embodiment of the present disclosure.

[0006] FIG. **3**A is a transverse cross-sectional view of the heat exchanger taken along line **3**A-**3**A in FIG. **1**, according to an embodiment of the present disclosure.

[0007] FIG. **3**B is an enlarged detailed transverse cross-sectional view of a section of the heat exchanger shown in FIG. **3**A, according to an embodiment of the present disclosure.

[0008] FIG. **4** is a schematic view of a section of the heat exchanger, according to an embodiment of the present disclosure.

[0009] FIG. **5** is a schematic representation of the particle separation system, according to an embodiment of the present disclosure.

[0010] FIG. **6** shows a simulated trajectory of airflow containing particles around the particle separation system and the heat exchanger, according to an embodiment of the present disclosure.

[0011] FIG. **7** is a schematic view of a particle separation system and heat exchanger shown in FIG. **2**A, according to an embodiment of the present disclosure.

[0012] FIGS. **8**A and **8**B are schematic views of the particle separation system and heat exchanger, according to embodiments of the present disclosure.

[0013] FIG. **9**A is a schematic view of the particle separation system and heat exchanger, according to another embodiment of the present disclosure.

[0014] FIG. **9**B is a schematic cross-sectional view of the heat exchanger taken at cross section line **9**B-**9**B in FIG. **9**A, according to an embodiment of the present disclosure.

[0015] FIG. **10** is a schematic cross-sectional view of a gas turbine engine having a particle separation system positioned upstream of a heat exchanger within a passage of a nacelle of the gas turbine engine such that incoming airflow outside of the nacelle enters the passage within the nacelle, according to another embodiment of the present disclosure.

[0016] FIG. **11** is a schematic cross-sectional view of a gas turbine engine having a particle separation system positioned upstream of a heat exchanger within a passage of the nacelle of the gas turbine engine such that incoming airflow inside of the nacelle enters the passage within the nacelle, according to another embodiment of the present disclosure.

[0017] FIG. **12** is a schematic cross-sectional view of a gas turbine engine having a particle separation system positioned upstream of a heat exchanger inside a passage within an outer casing of a turbo-engine of the gas turbine engine such that incoming airflow inside of the nacelle enters the passage of the outer casing of the turbo-engine, according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0018] Various embodiments are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the present disclosure.

[0019] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and the description have been used to refer to like or similar parts of the disclosure.

[0020] Generally, a turbofan engine includes a fan and a turbo-engine, with the turbo-engine rotating the fan to generate thrust. The turbo-engine includes a compressor section, a combustion section, a turbine section, and an exhaust section and defines a working gas flowpath therethrough.

[0021] Within the turbofan engine, thermal management is used to ensure the turbofan engine operates as desired or intended, without allowing certain components to exceed temperature thresholds. For example, a thermal management system of the turbofan engine may operate to reduce a temperature of a lubrication oil (used, e.g., for various bearings and gears within the turbofan engine), a cooled cooling air, electronics systems, etc., during operation of the turbofan engine. The thermal management system may utilize a thermal fluid and a heat exchanger to transfer heat from a heat source to a heat sink.

[0022] Generally, it is desirable to operate the thermal fluid and heat exchanger to enhance heat transfer properties of a thermal fluid to cool down a lubricant fluid (e.g., oil). However, when the heat exchanger is utilized within a working gas flow path of a turbofan engine, particles (e.g., dust particles, aerosol particles) within the gas flow (airflow) may create blockage of the air flow when passing through the heat exchanger. In addition, deposition of the particles on surfaces of the heat exchanger may be detrimental to heat transfer properties of the heat exchanger, as deposited dust particles have low thermal conductivity compared to metal.

[0023] The present disclosure provides a particle separator upstream of the heat exchanger heat exchanger, for example, upstream of plate-fin portions of the heat exchanger. The particle separator redirects dispersed particles (e.g., dust particles, aerosol particles) in the airflow to inner and/or outer passages of the heat exchanger so that the majority of a core of the heat exchanger remains particle-free. Redirecting the particles to inner and outer passages of the heat exchanger can provide many benefits, including:

[0024] (1) Flow into the heat exchanger may have various radial profiles resulting from upstream turbomachinery airfoils, variable geometry features, and change in engine operating power and/or airflow. Variation in flow velocity and pressure in the engine direction can result in the accompanying particles being directed to different radial locations of the heat exchanger.

[0025] (2) The inner and outer passages of the heat exchanger may not contribute to the total heat transfer as strongly as the central passages and thus can be used as a path to pass through the particles.

[0026] (3) Fins of the heat exchanger in inner and outer passages are in general shorter or have a shorter length than fins within the core of the heat exchanger (due to mechanical tolerances with respect to casing) and can have lower heat exchange effectiveness as the fins are connected to one plate rather than two.

[0027] The particle separator upstream and the heat exchanger heat exchanger can be provided in an annular configuration or a non-annular configuration, as will be described in further detail in the following paragraphs.

[0028] FIG. **1** is a schematic cross-sectional view of a gas turbine engine **100**, according to embodiment of the present disclosure. FIG. **1** shows the gas turbine engine **100** (e.g., a turbofan engine) having a rotor assembly with a single stage of unducted rotor blades. In such a manner, the rotor assembly may be referred to herein as an "unducted fan," or the entire gas turbine engine **100** may be referred to as an "unducted turbofan engine." In addition, the gas turbine engine **100** of FIG. **1** includes a first stream corresponding to a primary airflow passing through a fan **152**, a second stream flowing through a core duct **142** (e.g., a working gas flowpath through a turbo-engine **120**), and a third stream extending from the compressor section to a rotor assembly flowpath over the turbo-engine, as will be explained in more detail below. The "third stream" as used herein means a non-primary air stream capable of increasing fluid energy to produce a minority of total propulsion system thrust. The third stream may generally receive inlet air (air from a ducted passage downstream of a primary fan) instead of freestream air (as the primary fan would). A pressure ratio of the third stream may be higher than that of the primary propulsion stream (e.g., a bypass or a propeller driven propulsion stream). The thrust may be produced through a dedicated nozzle or through mixing of an airflow through the third stream with a primary propulsion stream or a core air stream, e.g., into a common nozzle.

[0029] The gas turbine engine **100** defines an axial direction A, a radial direction R, and a circumferential direction C. The gas turbine engine **100** defines an axial centerline or a longitudinal centerline axis **112** that extends along the axial direction A. The axial direction A extends parallel to the longitudinal centerline axis **112**. The radial direction R extends outward from and inward to the longitudinal centerline axis **112** in a direction orthogonal to the axial direction A. The circumferential direction extends three hundred sixty degrees around the axial direction A. The gas turbine engine **100** extends between a forward end **114** and an aft end **116**, e.g., along the axial direction A.

[0030] The gas turbine engine **100** includes a turbo-engine **120** and a rotor assembly or a fan section **150**, positioned upstream thereof. Generally, the turbo-engine **120** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. The turbo-engine **120** includes a core cowl **122** that defines an annular core inlet **124**. The core cowl **122** encloses at least in part a low pressure system and a high pressure system. For example, the core cowl **122** encloses and supports at least in part a booster or a low-pressure compressor (LPC) **126** for pressurizing the air that enters the turbo-engine **120** through the annular core inlet **124**. A high pressure, multi-stage, axial-flow compressor (HPC) **128** receives pressurized air from the LPC **126** and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor **130** of the combustion section where fuel is injected into the pressurized air stream and ignited to raise the temperature of the pressurized air.

[0031] The terms "high/low pressure" are relative terms and are used with respect to the LPC and HPC systems. Therefore, the terms "high" and "low" are used in this same context to distinguish the two systems, and are not meant to imply any absolute pressure values.

[0032] The high energy combustion products flow from the combustor **130** downstream to a high-pressure turbine (HPT) **132**. The HPT **132** drives the HPC **128** through a high-pressure shaft **136**. The HPT **132** is drivingly coupled with the HPC **128**. The hot combustion products then flow to a low-pressure turbine (LPT) **134**. The LPT **134** drives the LPC **126** and components of the fan section **150** through a low-pressure shaft **138**. The LPT **134** is drivingly coupled with the LPC **126** and components of the fan section **150**. In the present embodiment, the low-pressure shaft **138** is coaxial with the high-pressure shaft **136**. The combustion products that drive each of the HPT **132** and LPT **134** exit the turbo-engine **120** through a turbo-engine exhaust nozzle **140**.

[0033] Accordingly, the turbo-engine **120** defines a working gas flowpath or a core duct **142** that extends between the annular core inlet **124** and the turbo-engine exhaust nozzle **140**. The core duct **142** is an annular duct positioned generally inward of the core cowl **122** along the radial direction R. The core duct **142** (e.g., a working gas flowpath through the turbo-engine **120**) may be referred to as a second stream. The first stream corresponds to the primary airflow passing through the fan **152**.

[0034] The fan section **150** includes a fan **152**, which is the primary fan in this embodiment. For the embodiment of FIG. **1**, the fan **152** is an open rotor or an unducted fan **152**. In such a manner, the gas turbine engine **100** may be referred to as an open rotor or an unducted fan engine.

[0035] As depicted, the fan **152** includes a plurality of fan blades **154** (only one shown in FIG. **1**). The plurality of fan blades **154** are rotatable, e.g., about the longitudinal centerline axis **112**. As noted above, the fan **152** is drivingly coupled with the LPT **134** via the low-pressure shaft **138**. As shown in FIG. **1**, the fan **152** is coupled with the low-pressure shaft **138** via a speed reduction gearbox **155**, e.g., in an indirect-drive or a geared-drive configuration.

[0036] Moreover, the plurality of fan blades **154** can be arranged in equal spacing around the longitudinal centerline axis **112**. Each fan blade in the plurality of fan blades **154** has a root and a tip, and a span defined therebetween. Each fan blade in the plurality of fan blades **154** has a central blade axis **156**. Each fan blade in the plurality of fan blades **154** of the fan **152** is rotatable about its central blade axis **156**, e.g., in unison with one another. One or more actuators **158** are provided to facilitate such rotation and, therefore, may be used to change a pitch of the plurality of fan blades **154** about their respective central blades' axes **156**.

[0037] The fan section **150** further includes a fan guide vane array **160** that includes a plurality of fan guide vanes **162** (only one shown in FIG. **1**) disposed around the longitudinal centerline axis **112**. The plurality of fan guide vanes **162** are not rotatable about the longitudinal centerline axis **112**. Each fan guide vane in the plurality of fan guide vanes **162** has a root and a tip, and a span defined therebetween. The plurality of fan guide vanes **162** may be unshrouded as shown in FIG. **1** or, alternatively, may be shrouded, e.g., by an annular shroud spaced outward from the tips of the plurality of fan guide vanes **162** along the radial direction R or attached to the plurality of fan guide vanes **162**.

[0038] Each fan guide vane in the plurality of fan guide vanes **162** defines a central blade axis **164**. Each fan guide vane in the plurality of fan guide vanes **162** of the fan guide vane array **160** is rotatable about its respective central blade axis **164**, e.g., in unison with one another. One or more actuators **166** are provided to facilitate such rotation and, therefore, may be used to change a pitch of each fan guide vane in the plurality of fan guide vanes **162** about its respective central blade axis **164**. However, in other embodiments, each fan guide vane in the plurality of fan guide vanes **162** may be fixed or unable to be pitched about its central blade axis **164**. The plurality of fan guide vanes **162** are mounted to the fan cowl **170**. The fan cowl **170** is, generally, provided radially outer of the core cowl **122**.

[0039] As shown in FIG. **1**, in addition to the fan **152**, which is unducted, a ducted fan **184** is included aft of the fan **152**, such that the gas turbine engine **100** includes both a ducted fan and an unducted fan that both serve to generate thrust through the movement of air without passage through at least a portion of the turbo-engine **120** (e.g., without passage through the HPC **128** and combustion section in the depicted embodiment). The ducted fan **184** is rotatable about the same axis (e.g., the longitudinal centerline axis **112**) as the fan blade **154**. The ducted fan **184** is driven by the LPT **134** (e.g., coupled to the low-pressure shaft **138**). The unducted fan **152** may be referred to as the primary fan, and the ducted fan **184** may be referred to as a secondary fan. These terms “primary” and “secondary” are terms of convenience, and do not imply any particular importance, power, or the like.

[0040] The ducted fan **184** includes a plurality of fan blades (not separately labeled in FIG. **1**) arranged in a single stage, such that the ducted fan **184** may be referred to as a single stage fan. The fan blades of the ducted fan **184** can be arranged in equal spacing around the longitudinal centerline axis **112**. Each blade of the ducted fan **184** has a root and a tip, and a span defined therebetween.

[0041] The fan cowl **170** annularly encases at least a portion of the core cowl **122** and is generally positioned outward of at least a portion of the core cowl **122** along the radial direction R.

Particularly, a downstream section of the fan cowl **170** extends over a forward portion of the core cowl **122** to define a fan duct flowpath, or, simply, a fan duct **172**. According to this embodiment, the fan flowpath or fan duct **172** may form at least a portion of the third stream of the gas turbine engine **100** (e.g., another working gas flowpath of the gas turbine engine **100**).

[0042] Incoming air may enter through the fan duct **172** through a fan duct inlet **176** and may exit through a fan exhaust nozzle **178** to produce propulsive thrust. The fan duct **172** is an annular duct positioned generally outward of the core duct **142** along the radial direction R. The fan cowl **170** and the core cowl **122** are connected together and supported by a plurality of substantially radially-extending, stationary struts **174** that are circumferentially-spaced (only one shown in FIG. **1**). The stationary struts **174** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to the stationary struts **174** may be used to connect and to support the fan cowl **170** and/or the core cowl **122**. In many embodiments, the fan duct **172** and the core duct **142** may at least partially co-extend (generally axially) on opposite sides (e.g., opposite radial sides) of the core cowl **122**. For example, the fan duct **172** and the core duct **142** may each extend directly from a leading edge **144** of the core cowl **122** and may partially co-extend generally axially on opposite radial sides of the core cowl **122**.

[0043] The gas turbine engine **100** also defines or includes an inlet duct **180**. The inlet duct **180** extends between the engine inlet **182** and the annular core inlet **124** or fan duct inlet **176**. The engine inlet **182** is defined generally at the forward end of the fan cowl **170** and is positioned between the fan **152** and the fan guide vane array **160** along the axial direction A. The inlet duct **180** is an annular duct that is positioned inward of the fan cowl **170** along the radial direction R. Air flowing downstream along the inlet duct **180** is split, not necessarily evenly, into the core duct **142** and the fan duct **172** by a fan duct splitter or a leading edge **144** of the core cowl **122**. In the embodiment depicted, the inlet duct **180** is wider than the core duct **142** along the radial direction R. The inlet duct **180** is also wider than the fan duct **172** along the radial direction R.

[0044] The gas turbine engine **100** of FIG. **1** includes the first stream corresponding to a primary airflow passing through a fan **152**, the second stream flowing through a core duct **142** (e.g., a working gas flowpath through a turbo-engine **120**), and the third stream extending from the compressor section to a rotor assembly flowpath over the turbo-engine. The gas turbine engine **100** includes one or more features to increase an efficiency of the third stream thrust (e.g., a thrust generated by an airflow through the fan duct **172** exiting through the fan exhaust nozzle **178**, generated at least in part by the ducted fan **184**). The gas turbine engine **100** further includes an array of inlet guide vanes **186** positioned in the inlet duct **180** upstream of the ducted fan **184** and downstream of the engine inlet **182**. The array of inlet guide vanes **186** are arranged around the longitudinal centerline axis **112**. In this embodiment, the inlet guide vanes **186** are not rotatable about the longitudinal centerline axis **112**. Each of the inlet guide vanes **186** defines a central blade axis (not labeled for clarity), and is rotatable about its respective central blade axis, e.g., in unison with one another. In such a manner, the inlet guide vanes **186** may be considered a variable geometry component. One or more actuators **188** are provided to facilitate such rotation and, therefore, may be used to change a pitch of the inlet guide vanes **186** about their respective central blade axes. Alternatively, in other embodiments, each inlet guide vanes **186** may be fixed or unable to be pitched about its central blade axis.

[0045] The gas turbine engine **100** includes a plurality of outlet guide vanes **190** located downstream of the ducted fan **184** and upstream of the fan duct inlet **176**. As with the plurality of inlet guide vanes **186**, the plurality of outlet guide vanes **190** are not rotatable about the longitudinal centerline axis **112**. However, in the embodiment depicted in FIG. **1**, unlike the plurality of inlet guide vanes **186**, the plurality of outlet guide vanes **190** are configured as fixed-pitch outlet guide vanes.

[0046] The fan exhaust nozzle **178** of the fan duct **172** is further configured as a variable geometry exhaust nozzle. In such a manner, the gas turbine engine **100** includes one or more actuators **192** for modulating the fan exhaust nozzle **178**. For example, the fan exhaust nozzle **178** may be configured to vary a total cross-sectional area (e.g., an area of the nozzle in a plane perpendicular to the longitudinal centerline axis **112**) to modulate an amount of thrust generated based on one or more engine operating conditions (e.g., temperature, pressure, mass flowrate, etc., of an airflow through the fan duct **172**). A fixed geometry exhaust nozzle may also be adopted.

[0047] The combination of the plurality of inlet guide vanes **186** located upstream of the ducted fan **184**, the plurality of outlet guide vanes **190** located downstream of the ducted fan **184**, and the fan exhaust nozzle **178** may result in a more efficient generation of third stream thrust during one or more engine operating conditions. In addition, by introducing a variability in the geometry of the inlet guide vanes **186** and the fan exhaust nozzle **178**, the gas turbine engine **100** may be capable of generating more efficient third stream thrust across a relatively wide array of engine operating conditions, including takeoff and climb (when a higher total engine thrust, is generally needed) as well as cruise (when a lesser amount of total engine thrust, is generally needed).

[0048] Moreover, referring still to FIG. **1**, in exemplary embodiments, air passing through the fan duct **172** may be relatively cooler (e.g., at a lower temperature) than one or more fluids utilized in the turbo-engine **120**. In this way, one or more heat exchangers **198** are positioned in thermal communication with the fan duct **172**. For example, one or more heat exchanger **198** are disposed within the fan duct **172** and utilized to cool one or more fluids from the core engine with the air passing through the fan duct **172**, as a resource for removing heat from the one or more fluids (e.g., compressor bleed air, oil, fuel, etc.).

[0049] Although not depicted, the heat exchanger **198** may be an annular heat exchanger extending substantially three hundred sixty (360) degrees in the fan duct **172** (e.g., at least three hundred (300) degrees, such as at least three hundred thirty (330) degrees). In such a manner, the heat exchanger **198** may effectively utilize the air passing through the fan duct **172** to cool one or more systems of the gas turbine engine **100** (e.g., lubrication oil systems, compressor bleed air, electrical components, etc.). The heat exchanger **198** uses the air passing through the fan duct **172** as a heat sink and correspondingly increases the temperature of the air downstream of the heat exchanger **198** and exiting the fan exhaust nozzle **178**.

[0050] The gas turbine engine **100** in FIG. **1** is depicted by way of example only. In other exemplary embodiments, the gas turbine engine **100** may have any other suitable configuration. For example, in other embodiments, the gas turbine engine **100** may be configured as a ducted gas turbine engine (e.g., including a nacelle surrounding the fan **152**), may be a direct drive gas turbine engine (e.g., driving the fan **152** without use of a speed reduction gearbox **155**), etc.

[0051] In addition, although the heat exchanger **198** is depicted positioned in the fan duct **172**, in other exemplary embodiments, the gas turbine engine **100** may additionally or alternatively include a heat exchanger **199** in the turbo-engine exhaust nozzle **140**, or at other suitable or desired locations.

[0052] The gas turbine engine **100** also includes a particle separation system **197** disposed upstream of the heat exchanger **198**. The heat exchanger **198** and the particle separation system **197** are provided in an annular configuration where both the heat exchanger **198** and the particle separation system **197** are annular and located at a radial position relative to the longitudinal centerline axis **112** and extend circumferentially in the circumferential direction C around the longitudinal centerline axis **112**. The structure and function of the particle separation system **197** will be described in further detail in the following paragraphs.

[0053] FIG. **2**A shows a location of a heat exchanger **200** along with a particle separation system **204** within the fan duct **172** of the gas turbine engine **100**, according to an embodiment of the present disclosure. The particle separation system **204** can be used as the particle separation system **197** in the gas turbine engine **100**. In an embodiment, the particle separation system **204** includes one or more louvers **203** (one or more inertial particle separator). As described in the above paragraphs, the heat exchanger **200** can be positioned in thermal communication with the fan duct

172. The heat exchanger **200** disposed within the fan duct **172** is utilized to cool down one or more fluids **206** from the core engine with incoming airflow **208** passing through the fan duct **172**, as a resource for removing heat from the one or more fluids **206** (e.g., compressor bleed air, oil, fuel, etc.). The one or more fluids **206** (e.g., oil) circulates within the heat exchanger **200** and the incoming airflow **208** passing through the heat exchanger **200**, **202** cools down the one or more fluids **206** (e.g., oil). A temperature of the one or more fluids **206** is greater than a temperature of the incoming airflow **208**. The incoming airflow **208** is a portion of airflow **210** passing through the engine inlet **182** (shown in FIG. **1**). The heat exchanger **200** can be similar to the heat exchanger **198** shown in FIG. **1**.

[0054] In an embodiment, the particle separation system **204** can be provided upstream of the heat exchanger **200**. In an embodiment, the particle separation system **204** is configured to divide the incoming airflow **208** incident on the particle separation system **204** into a first airflow **208**A containing a first amount of particles and a second airflow **208**B containing substantially no particles or a second amount of particles that is substantially less than the first amount of particles. The particle separation system **204** is configured to direct the first airflow **208**A towards a radial periphery of the fan duct **172** so that the particles in the first airflow **208**A pass through edges of the heat exchanger **202** substantially unhindered by the heat exchanger **200** and the second airflow **208**B containing substantially less or no particles pass through a central portion of the heat exchanger **200** to cool down the one or more fluids **206** circulating within the heat exchanger **200**. As a result, a majority of particles are separated from the incoming airflow **208** to obtain the second airflow **208**B containing substantially fewer particles. In an embodiment, the particle separation system **204** is a mechanical device, having, for example, one or more louvers **203** that are configured and arranged to deviate the particles in the incoming airflow **208**. An airflow with “substantially no or fewer particles” is to be understood as an airflow containing “less than about 80%” of an original concentration of particles in the incoming airflow **208**. “Concentration of particles” is understood to mean either mass or volume concentration of particles. The first amount of particles in the first airflow **208**A may have be a concentration of particles greater than about 80% of a concentration of particles in the incoming airflow **208** while the second amount of particles in the second airflow **208**B may have a concentration of particles less than about 20% of the of the original concentration of particles in the incoming airflow **208**. The first amount of particles (with a concentration of particles greater than 80%) is substantially greater than the second amount of particles (with a concentration of particles less than 20%).

[0055] The particles may include, but not are not limited to, aerosol particles, sand particles, ice particles, dust, foreign object debris (FOD), etc. The particles can have an average size between about one (1) micron and about one (1) millimeter (e.g., between about 5 microns and about one hundred (100) microns).

[0056] FIG. **2**B shows a location of a heat exchanger **202** along with a particle separation system **205** within the fan duct **172** of the gas turbine engine **100**, according to another embodiment of the present disclosure. The particle separation system **205** can be also used as the particle separation system **197** in the gas turbine engine **100**. The particle separation system **205** (shown in FIG. **2**B) can be used as the particle separation system **197** in the gas turbine engine **100** instead of, or in addition to, particle separation system **204** (shown in FIG. **2**A). FIG. **2**B illustrates an upstream conditioning device to direct particles towards outer strata of the flow path so the present particle separation system may be positioned accordingly to further separate those particles. Conversely, the present particle separation system may be positioned upstream of another particle separation system as a pre-conditioner. For example, the present device may be positioned as shown in FIG. **2**A to deflect particles to the inner and outer edges of the flow path where filters are positioned to trap the particles. As described in the above paragraphs, the heat exchanger **202** can be positioned in thermal communication with the fan duct **172**. The heat exchanger **202** disposed within the fan duct **172** is utilized to cool down one or more fluids **206** from the core engine with incoming airflow **208** passing through the fan duct **172**, as a resource for removing heat from the one or more fluids **206** (e.g., compressor bleed air, oil, fuel, etc.). The one or more fluids **206** (e.g., oil) circulates within the heat exchanger **200** and the incoming airflow **208** passing through the heat exchanger **202** cools down the one or more fluids **206** (e.g., oil). A temperature of the one or more fluids **206** is greater than a temperature of the incoming airflow **208**. The incoming airflow **208** is a portion of airflow **210** passing through the engine inlet **182** (shown in FIG. **1**). The heat exchanger **202** can be similar to the heat exchanger **198** shown in FIG. **1**.

[0057] In an embodiment, the particle separation system **205** can be provided upstream of the heat exchanger **202**. In an embodiment, the particle separation system **205** is configured to divide the incoming airflow **208** incident on the particle separation system **205** into a first airflow **208**A containing a first amount of particles and a second airflow **208**B containing substantially no particles or a second amount of particles that is substantially less than the first amount of particles.

[0058] In an embodiment, the particle separation system **205** is configured to deviate the first airflow **208**A containing a major amount of particles towards a wall of the fan duct **172** so that the particles pass through edges of the heat exchanger **202** substantially unhindered by the heat exchanger **202** and to direct the second airflow **208**B containing substantially fewer particles to pass through a central portion of the heat exchanger **202** to cool down the one or more fluids **206** circulating within the heat exchanger **202**. In an embodiment, the particle separation system **205** can take the form of a shaped turn surface **172**A (inertial particle separator) of the fan duct **172**. The shaped turn surface **172**A is configured and arranged so that the first airflow containing the majority of particles, and having a relatively higher momentum than the second airflow **208**B having fewer or no particles, collide with the shaped turn surface **172**A to deviate the first airflow **208**A towards a radial periphery of the fan duct **172** and, thus, to the radial periphery of the heat exchanger **202**. On the other hand, the second airflow **208**B containing fewer or no particles does not collide with the shaped turn surface **172**A and, thus, is not deviated and passes through a central portion of the heat exchanger **202**. In an embodiment, due to the presence of a sharp turn in the flow path, particles that are more inertial than fluid particles, deviate from the flow streamlines and are thrown to the outer passage of the heat exchanger. In an embodiment, the heat exchanger **200** and the particle separation system **204** are provided in an annular configuration where both the heat exchanger **200** and the particle separation system **204** are annular and located at a radial position relative to the longitudinal centerline axis **112** and extend circumferentially in the circumferential direction C around the longitudinal centerline axis **112**. In another embodiment, the heat exchanger **200** along with the particle separation system **204** can also be configured for a non-annular passage such as an offtake branch of a main airflow passage.

[0059] FIG. **3**A is a transverse cross-sectional view of the heat exchanger **200** and heat exchanger **202** taken along line **3**A-**3**A of FIG. **1**, FIGS. **2**A, and FIG. **2**B parallel to the radial direction R and perpendicular to the axial direction A (shown in FIG. **1**), according to an embodiment of the present disclosure. As shown in FIG. **3**A, the heat exchanger **200** and heat exchanger **202** have an annular cross section that corresponds to the annular shape of the fan duct **172**. The fan duct **172** is an annular duct positioned generally outward of the core duct **142** (shown in FIG. **1**) along the radial direction R.

[0060] FIG. **3**B is an enlarged detailed transverse cross-sectional view of a section **250** of the heat exchanger **200** and heat exchanger **202** shown in FIG. **3**A, according to an embodiment of the present disclosure. As shown in FIG. **3**B, the heat exchanger **200** and heat exchanger **202** include a plurality of annular plates **300**. The plurality of annular plates **300** are spaced apart from each other and are distributed between a hub **302** located at an inner most radial position R**1** and a tip **304** located at an outer most radial position R**2**. The heat exchanger **200** and heat exchanger **202** also include a plurality of fins **306**. The plurality of fins **306** include a first plurality of fins **306**A provided between the plurality of annular plates **300**. In an embodiment, the first plurality of fins **306**A are circumferentially spaced apart (e.g., spaced apart along the circumference C) and are coupled to the plurality of annular plates **300**. The second plurality of fins **306**B are located between one of the plurality of annular plates **300** and the hub **302** and between another one of the plurality of annular plates **300** and the tip **304**. In an embodiment, the second plurality of fins **306**B are also circumferentially spaced apart. The second plurality of fins **306**B are connected to or coupled to one of the plurality of annular plates **300**. The second plurality of fins **306**B are not connected or coupled to the hub **302** to provide an inner passage **302**A. The second plurality of fins **306**B are also not coupled to the tip **304** to provide an outer passage **304**A. A length of second plurality of fins **306**B is less than a length of the first plurality of fins **306**A. In an embodiment, the length of second plurality of fins **306**B is less than a length of the first plurality of fins **306**A and the second plurality of fins **306**B are only connected at one end to one of the plurality of annular plates **300** so as to allow the first airflow **208**A (shown in FIG. **2**) containing the majority of particles to pass therethrough. In an embodiment, the first plurality of fins **306**A and/or the second plurality of fins **306**B can be arranged substantially parallel to each other, as shown in FIG. **3**B. In another embodiment, the first plurality of fins **306**A and/or the second plurality of fins **306**B can be arranged in a "V" configuration or wave configuration, as shown in FIG. **4**.

[0061] In an embodiment, a number of the second plurality of fins **306**B per inch provided at the hub **302** and at the tip **304** of the heat exchanger **200** can be adjusted in the inner passage **302**A at the hub **302** and the outer passage **304**A at the tip **304** of the heat exchanger **200** to prevent blockage and mitigate particles deposition.

[0062] FIG. **4** is a schematic perspective view of a section of the heat exchanger **200** or heat exchanger **202**, according to an embodiment of the present disclosure. The heat exchanger **200** or heat exchanger **202** includes the plurality of annular plates **300** that are spaced apart from each other. Each of the plurality of annular plates **300** includes one or more channels **300**A. The one or more channels **300**A are used to circulate a fluid **309** (e.g., oil used for lubrication). The one or more channels **300**A can be a plurality of channels or a single serpentine channel within each of the plurality of annular plates **300**. As shown in FIG. **4**, the plurality of fins **306** can be arranged parallel to each other, as shown, for example, at the upper section of the heat exchanger **200** or heat exchanger **202**, or the plurality of fins **306** can also be arranged in a "V" configuration, as shown in the middle section of the heat exchanger **200** or heat exchanger **202**, and/or the plurality of fins **306** can further be arranged in a wave configuration, as shown in the lower section of the heat exchanger **200** or heat exchanger **202**. As shown in FIG. **4**, an airflow **310** (i.e., a flow of cold fluid) is introduced between the plurality of annular plates **300** and between the plurality of fins **306** to cool down the fluid **309** that circulates within the one or more channels **300**A of each of the plurality of annular plates **300**. The terms "hot" and "cold" are relative terms and are used to mean that a temperature of the fluid **309** (e.g., hot fluid) is greater than a temperature of the airflow **310** (cold fluid).

[0063] FIG. **5** is a schematic representation of the particle separation system **204**, according to an embodiment of the present disclosure. As shown in FIG. **5**, the particle separation system **204** includes one or more louvers **500**. The one or more louvers **500** are arranged in an alternating configuration such that a first louver **500**A is oriented at a positive angle α relative to the axial direction A (i.e., axial direction of the heat exchanger **200**), a second louver **500**B is oriented at a negative angle α relative to the axial direction, a third louver **500**C is oriented at a positive angle θ relative to the axial direction A, and a fourth louver **500**D is oriented at a negative angle θ relative to the axial direction A, etc. In FIG. **5**, four louvers **500** are shown. However, any number of louvers **500** can be used, for example, one, two, three, or more louvers can be used. In an embodiment, an upstream edge of the second louver **500**B is spaced apart from an upstream edge of the first louver **500**A by a distance L**2**, an upstream edge of the third louver **500**C is spaced apart from a downstream edge of the first louver **500**A by a distance L**1**, and a downstream edge of the third louver **500**C is spaced apart from an upstream edge of the heat exchanger **200** by a distance L**3**. In an embodiment, the angles α and/or θ, the distances L**1**, L**2** and/or L**3** can be adjusted to enhance efficiency of separating the particles.

[0064] The one or more louvers **500** can be mounted to the heat exchanger **200** or mounted to the fan duct **172** (shown in FIGS. **1** and **2**). An actuation system **510** can be provided to rotate the one or more louvers **500** (one or more inertial particle separators). In an embodiment, the angles of the louvers can be adjusted and controlled throughout the flight mission as desired. For example, the angles α and θ can be set to zero degree when fewer particles are present to minimize or reduce pressure drop.

[0065] FIG. **6** shows a simulated trajectory of airflow containing particles around the particle separation system **204** and the heat exchanger **200**, according to an embodiment of the present disclosure. As shown in FIG. **6**, the incoming airflow **208** containing both particles are thrown and deviated by the particle separation system **204**. The particle separation system **204** is configured to deviate the first airflow **208**A containing a majority of particles towards a radial periphery of the fan duct **172** so that particles in the first airflow **208**A pass through edges of the heat exchanger **200** substantially unhindered by the heat exchanger **200** and direct the second airflow **208**B containing substantially fewer or no particles to pass through a central portion of the heat exchanger **200** to cool down the fluid **309** circulating within the heat exchanger **200**. As a result, the first airflow **208**A containing the majority of particles is separated from the second airflow **208**B containing substantially less particles. As a result, the particles in the first airflow **208**A is deviated by the louvers **500** towards a radial periphery of the fan duct **172** and, thus, to radial periphery of the heat exchanger **200** (e.g., towards the hub **302** and the tip **304** of the heat exchanger **200**). On the other hand, the second airflow **208**B containing less or no particles, and having less momentum than the airflow containing the particles **208**A, is not deviated by the louvers **500** and passes through a central portion of the particle separation system **204** and the central portion of the heat exchanger **200** to cool down the fluid **309** (shown in FIG. **4**) circulating inside the heat exchanger **200**.

[0066] FIG. **7** is a schematic view of a particle separation system **700** and heat exchanger **200** shown in FIG. **2**A and FIGS. **5** and **6**, according to an embodiment of the present disclosure. As shown in FIG. **7**, the heat exchanger **200** includes the plurality of annular plates **300** that are spaced apart from each other. Each of the plurality of annular plates **300** includes the one or more channels **300**A. The one or more channels **300**A are used to circulate a fluid **309**. For example, the fluid **309** can be a hot fluid such as an oil used for lubrication. In addition, the particle separation system **700** includes one or more louvers **702**. Each of the one or more louvers **702** also includes one or more passages **702**A provided to also circulate the fluid **309**. In this embodiment, the one or more louvers **702** of the particle separation system **700** can also play the role of a heat exchanger as the incoming airflow **208** can also be used to heat or cool down the fluid **309** circulating inside the one or more passages **702**A of the one or more louvers **702**. In addition, the one or more louvers **702** of the particle separation system **700** can also be used as an ice screen by melting ice particles in incoming airflow **208** with the fluid **309** to limit exposure of the heat exchanger **200** to ice particles. In an embodiment, the one or more louvers **702** (one or more particle separators) can be integrally manufactured (e.g., additively manufactured integral with the heat exchanger **200**), mounted to the heat exchanger **200** or mounted to the fan duct **172** (shown in FIGS. **1** and **2**). In another embodiment, the one or more louvers **702** can also be integrally manufactured with the heat exchanger **200** (e.g., additively manufactured to be integral with heat exchanger **200**). An actuation system **710** can be provided to rotate the one or more louvers **702**, for example, when the one or more louvers **702** are not rigidly fixed to the heat exchanger **200**.

[0067] In an embodiment, the one or more louvers **500** (shown in FIGS. **5** and **6**) and the one or more louvers **702** (shown in FIG. **7**) can also be used as a foreign object damage (FOD) or own/domestic object damage (OOD or DOD) shield to mitigate the impact of foreign object debris (FOD) (e.g., ice crystals, bird parts, snarge, or remains of a bird after it has collided with a turbine engine) on the heat exchanger **200** and/or to mitigate the impact of OOD (e.g., debris originating from the turbine engine itself such as blade chip fragments or the like) on the heat exchanger **200**.

For example, the FOD can be directed around the heat exchanger **200** to where blockage is less impactful on the efficiency of the heat exchanger **200**.

[0068] In an embodiment, the one or more louvers **500** (shown in FIGS. **5** and **6**) and the one or more louvers **702** (shown in FIG. **7**) can be integrated at the entrance of the heat exchanger **200** so that the energy loss that results from turning the primary airflow, can be transferred to some working fluid. The working fluid can be the same as the fluid **309** in heat exchanger **200** or part of a separate thermal transport bus. For example, when the one or more louvers **702** are integrated with the heat exchanger **200**, in addition to heat transfer benefits provided by the heat exchanger **200**, FOD, OOD, and/or ice particles, etc. screening benefits are also provided by the one or more louvers **702**.

[0069] FIGS. **8**A and **8**B are schematic views of a particle separation system **800** and the heat exchanger **200**, according to embodiments of the present disclosure. As shown in FIGS. **8**A and **8**B, the particle separation system **800** includes at least one inertial particle separator **802** or at least one inertial separator **804**. The at least one inertial particle separator **802** can be a mass having an oval or rounded shape. The at least one inertial separator **804** can have a triangular shape or pointed shape (e.g., a wedge). The at least one inertial separator can also be one or more louvers as described in the above paragraphs.

[0070] FIG. **9**A is a schematic view of a particle separation system **900** and a heat exchanger **910**, according to another embodiment of the present disclosure. As shown in FIG. **9**A, the particle separation system **900** is a cyclonic separator. The particle separation system **900** is provided upstream of the heat exchanger **910**. The particle separation system **900** receives an incoming airflow **208**. The particle separation system **900** includes a housing **902** defining a circular interior **904** (e.g., circular duct). The particle separation system **900** includes a vortex generator **906** (inertial particle separator) disposed within the housing **902**. The vortex generator **906** includes a spiral coil **906**A coupled to a shaft **906**B. The shaft **906**B can be configured to rotate thus rotating the spiral coil **906**A. The rotation of the vortex generator **906** moves the particles in the incoming airflow **208** towards the radial outer periphery of the housing **902** by a centrifugal force generated by the rotation of the vortex generator **906** to provide the first airflow **208**A containing a first amount of particles. The first airflow **208**A containing the first amount of particles travels along the radial outer periphery of the housing **902** and then passes into the radially outer volume of the fan duct **172**.

[0071] FIG. **9**B is a schematic cross-sectional view of the heat exchanger **910** taken at cross section line **9**B-**9**B in FIG. **9**A, according to an embodiment of the present disclosure. As shown in FIG. **9**B, the first airflow **208**A further passes through peripheral channel **302**B of the heat exchanger **910**. As shown in FIG. **9**B, the heat exchanger **910** has the same housing **902** as the particle separation system **900**. The heat exchanger **910** also has a plurality of circular plates **912** connected to each other via a plurality fins **914**. The plurality of circular plates **912** are spaced apart radially from each other. The plurality of fins **914** are arranged radially and are circumferentially spaced apart (e.g., spaced apart along the circumference C) and are coupled to the plurality of circular plates **912**. In an embodiment, at least a portion of the plurality of fins **914** are not coupled to the housing **902** to define the peripheral channel **302**B. As shown in FIG. **9**A, the first airflow **208**A having the first amount of particles travels along the radial outer periphery of the housing **902** and passes through the peripheral channel **302**B. As shown in FIG. **9**A, the second airflow **208**B having a second amount of particles (e.g., the airflow with substantially fewer particles or no particles) is substantially not affected by the centrifugal force generated by the rotation of the vortex generator **906** and is allowed to pass through the vortex generator **906**. As a result, the second airflow **208**B continues towards the central portion of the heat exchanger **910** to cool down the fluid **309** circulating with the heat exchanger **910**. The second amount of particles is substantially less than the first amount of particles. The first amount of particles in the first airflow **208**A may have a concentration of particles greater than about 80% of an original concentration of particles in the incoming airflow **208** while the second amount of particles in the second airflow **208**B may have a concentration of particles less than about 20% of the original concentration of particles in the incoming airflow **208**. The first amount of particles (with a concentration of particles greater than 80%) is, therefore, substantially greater than the second amount of particles (with a concentration of particles less than 20%).

[0072] The particle separation system described above redirects dispersed particles (e.g., dust particles, aerosol particles) in the airflow to inner and/or outer passages of the heat exchanger so that the majority of a core of the heat exchanger remains particle-free. Redirecting the particles to inner and outer passages of the heat exchanger can provide many benefits, including:

[0073] (1) Directing particles to the generally lower flow regions of the approach airflow for bypassing the heat exchanger does not significantly affect the heat exchanger thermal performance because a majority of the airflow is flowing through the radial central portion of the duct. The outer passages of the heat exchanger participate less strongly in heat transfer than the central passages of the heat exchanger due to the radial flow profile.

[0074] (2) Fins of the heat exchanger in inner and outer passages are in general shorter or have a smaller length than fins within the core of the heat exchanger (due to mechanical tolerances with respect to casing) and can have lower heat exchange effectiveness as the fins are connected to one plate rather than two.

[0075] FIG. **10** is a schematic cross-sectional diagram of a gas turbine engine **1000**, according to another embodiment of the present disclosure. As shown in FIG. **10**, the gas turbine engine **1000** defines an axial direction A (extending parallel to a longitudinal centerline **1012** provided for reference) and a radial direction R that is normal to the axial direction A. In general, the gas turbine engine **1000** includes a fan section **1014** and a turbo-engine **1016** disposed downstream from the fan section **1014**.

[0076] The turbo-engine **1016** depicted generally includes an outer casing **1018** that is substantially tubular and defines an annular inlet **1020**. As schematically shown in FIG. **10**, the outer casing **1018** encases, in serial flow relationship, a compressor section **1021** including a booster or a low pressure (LP) compressor **1022** followed downstream by a high pressure (HP) compressor **1024**, a combustion section **1026**, a turbine section **1027** including a high pressure (HP) turbine **1028** followed downstream by a low pressure (LP) turbine **1030**, and a jet exhaust nozzle section **1032**. A high pressure (HP) shaft or spool **1034** drivingly connects the HP turbine **1028** to the HP compressor **1-24** to rotate the HP turbine **1028** and the HP compressor in unison. A low pressure (LP) shaft **1036** drivingly connects the LP turbine **1030** to the LP compressor **1022** to rotate the LP turbine **1030** and the LP compressor **1022** in unison. The compressor section **1021**, the combustion section **1026**, the turbine section **1027**, and the jet exhaust nozzle section **1032** together define a core air flowpath.

[0077] For the embodiment depicted in FIG. **10**, the fan section **1014** includes a fan **1038** (e.g., a variable pitch fan) having a plurality of fan blades **1040** coupled to a disk **1042** in a spaced apart manner. As depicted in FIG. **10**, the fan blades **1040** extend outwardly from the disk **1042** generally along the radial direction R. Each fan blade **1040** is rotatable relative to the disk **1042** about a pitch axis P by virtue of the fan blades **1040** being operatively coupled to an actuation member **1044** configured to collectively vary the pitch of the fan blades **1040** in unison. The fan blades **1040**, the disk **1042**, and the actuation member **1044** are together rotatable about the longitudinal centerline **1012** via a fan shaft **1045** that is powered by the LP shaft **1036** across a power gearbox **1046**. The power gearbox **1046** includes a plurality of gears for adjusting the rotational speed of the fan shaft **1045** and, thus, the fan **1038** relative to the LP shaft **1036** to a more efficient rotational fan speed.

[0078] Referring still to the exemplary embodiment of FIG. **10**, the disk **1042** is covered by a rotatable fan hub **1048** aerodynamically contoured to promote an airflow through the plurality of fan blades **1040**. In addition, the fan section **1014** includes an annular fan casing or a nacelle **1050** that circumferentially surrounds the fan **1038** and/or at least a portion of the turbo-engine **16**. The nacelle **1050** is supported relative to the turbo-engine **1016** by a plurality of circumferentially spaced outlet guide vanes **1052**. Moreover, a downstream section **1054** of the nacelle **1050** extends over an outer portion of the turbo-engine **1016** to define a bypass airflow passage **1056** therebetween.

[0079] During operation of the gas turbine engine **1000** a volume of air **1058** enters the gas turbine engine **1000** through an inlet **1060** of the nacelle **1050** and/or the fan section **1014**. As the volume of air **1058** passes across the fan blades **1040**, a first portion of the air **1062** is directed or routed into the bypass airflow passage **1056**, and a second portion of the air **1064** is directed or is routed into the upstream section of the core air flowpath, or, more specifically, into the annular inlet **1020** of the LP compressor **1022**. The ratio between the first portion of air **1062** and the second portion of air **1064** is commonly known as a bypass ratio. The pressure of the second portion of air **1064** is then increased as it is routed through the HP compressor **1024** and into the combustion section **1026**, where the highly pressurized air is mixed with fuel and burned to provide combustion gases **1066**.

[0080] The combustion gases **1066** are routed into the HP turbine **1028** and expanded through the HP turbine **1028** where a portion of thermal and/or of kinetic energy from the combustion gases **1066** is extracted via sequential stages of HP turbine stator vanes **1068** that are coupled to the outer casing **1018** and HP turbine rotor blades **1070** that are coupled to the HP shaft or spool **1034**, thus causing the HP shaft or the spool **1034** to rotate, thereby supporting operation of the HP compressor **1024**. The combustion gases **1066** are then routed into the LP turbine **1030** and expanded through the LP turbine **1030**. Here, a second portion of thermal and kinetic energy is extracted from the combustion gases **1066** via sequential stages of LP turbine stator vanes **1072** that are coupled to the outer casing **1018** and LP turbine rotor blades **1074** that are coupled to the LP shaft **1036**, thus, causing the LP shaft **1036** to rotate. This thereby supports operation of the LP compressor **1022** and rotation of the fan **1038** via the power gearbox **1046**.

[0081] The combustion gases **1066** are subsequently routed through the jet exhaust nozzle section **1032** of the turbo-engine **1016** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **1062** is substantially increased as the first portion of air **1062** is routed through the bypass airflow passage **1056** before being exhausted from a fan nozzle exhaust section **1076** of the gas turbine engine **1000**, also providing propulsive thrust. The HP turbine **1028**, the LP turbine **1030**, and the jet exhaust nozzle section **1032** at least partially define a hot gas path **1078** for routing the combustion gases **1066** through the turbo-engine **1016**.

[0082] The gas turbine engine **1000** depicted in FIG. **10** is by way of example only. In other exemplary embodiments, the gas turbine engine **1000** may have any other suitable configuration. For example, in other exemplary embodiments, the fan **1038** may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Moreover, it should be appreciated that, in other exemplary embodiments, any other suitable number or configuration of compressors, turbines, shafts, or a combination thereof may be provided. In still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable gas turbine engine, such as, for example, turbofan engines, propfan engines, turbojet engines, and/or turboshaft engines.

[0083] In an embodiment, similar to the embodiment shown in FIG. **1**, the gas turbine engine **1000** also includes a particle separation system **1090** disposed upstream of the heat exchanger **1092**. The particle separation system **1090** can be similar to the particle separation system **197**, **204**, **205**, **700**, **800**, **802**, **804**, **900** described above. The heat exchanger **1092** can be similar to the heat exchanger **198**, **199**, **200**, **202** described above. While in the embodiment shown in FIG. **1**, the heat exchanger **198** and the particle separation system **197** are provided in an annular configuration, in the embodiment shown in FIG. **10**, the heat exchanger **1092** and the particle separation system **1090** are provided in a non-annular configuration. In FIG. **1**, the heat exchanger **198** and the particle separation system **197** are annular and located at a radial position relative to the longitudinal centerline axis **112** and extend circumferentially in the circumferential direction around the longitudinal centerline axis **112**. Whereas, in FIG. **10**, the particle separation system **1090** and the heat exchanger **1092** are located at a radial position relative to the longitudinal centerline axis **112** but do not extend circumferentially in the circumferential direction around the longitudinal centerline axis **112**.

[0084] As shown in FIG. **10**, the particle separation system **1090** and the heat exchanger **1092** are provided within the nacelle **1050** (i.e., the nacelle wall) and can be located at any position within the circumferential direction. The particle separation system **1090** is disposed upstream of the heat exchanger **1092**. The particle separation system **1090** and the heat exchanger **1092** are positioned within a passage **1051** of the nacelle **1050** such that incoming airflow **1091** outside of the nacelle **1050** enters the passage **1051** within the nacelle **1050** to first encounter the particle separation system **1090**. The particle separation system **1090** is configured to separate particles present in the incoming airflow **1091** to provide cleaner airflow (airflow containing substantially less particles). The cleaner airflow then continues towards the heat exchanger **1092** to cool down a fluid (not shown in FIG. **10**) circulating within the heat exchanger **1092** before exiting as output airflow **1093** to outside of the nacelle **1050** (i.e., outside of the wall of the nacelle **1050**).

[0085] FIG. **11** is a schematic cross-sectional view of a gas turbine engine, according to another embodiment of the present disclosure. In an embodiment, similar to the embodiment shown in FIG. **10**, the gas turbine engine **1001** also includes the particle separation system **1090** disposed upstream of the heat exchanger **1092**. The particle separation system **1090** can be similar to the particle separation system **197**, **204**, **205**, **700**, **800**, **802**, **804**, **900** described above. The heat exchanger **1092** can be similar to the heat exchanger **198**, **199**, **200**, **202** described above. As shown in FIG. **11**, similar to the embodiment shown in FIG. **10**, the particle separation system **1090** and the heat exchanger **1092** are provided within the nacelle **1050** (i.e., with the wall of the nacelle **1050**). Similar to the embodiment shown in FIG. **10**, the particle separation system **1090** and the heat exchanger **1092** are provided in a non-annular configuration wherein the particle separation system **1090** and the heat exchanger **1092** are located at a radial position relative to the longitudinal centerline axis **112** but do not extend circumferentially in the circumferential direction around the longitudinal centerline axis **112**. The particle separation system **1090** and the heat exchanger **1092** are provided within the nacelle **1050** (i.e., the nacelle wall) and can be located at any position within the circumferential direction.

[0086] The particle separation system **1090** is disposed upstream of the heat exchanger **1092**. The particle separation system **1090** and the heat exchanger **1092** are positioned within a passage **1053** of the nacelle **1050** (within the wall of the nacelle) such that incoming airflow **1095** inside of the nacelle **1050** enters the passage **1053** within the nacelle **1050** to first encounter the particle separation system **1090**. The particle separation system **1090** is configured to separate particles present in the incoming airflow **1091** to provide cleaner airflow (airflow having substantially less particles). The cleaner airflow then continues towards the heat exchanger **1092** to cool down a fluid (not shown in in FIG. **11**) circulating within the heat exchanger **1092** before exiting as output airflow **1097** to inside of the nacelle **1050**. The incoming airflow **1095** can be part of the first portion of air **1062**.

[0087] FIG. **12** is a schematic cross-sectional view of a gas turbine engine, according to another embodiment of the present disclosure. In an embodiment, similar to the embodiment shown in FIG. **1**, the gas turbine engine **1003** also includes the particle separation system **1090** disposed upstream of the heat exchanger **1092**. The particle separation system **1090** can be similar to the particle separation system **197**, **204**, **205**, **700**, **800**, **802**, **804**, **900** described above. The heat exchanger **1092** can be similar to the heat exchanger **198**, **199**, **200**, **202** described above. As shown in FIG. **12**, the particle separation system **1090** and the heat exchanger **1092** are provided within the outer casing **1018** of the turbo-engine **1016**. Similar to the embodiment shown in FIGS. **10** and **11**, the particle separation system **1090** and the heat exchanger **1092** are provided in a non-annular configuration wherein the particle separation system **1090** and the heat exchanger **1092** are located at a radial position relative to the longitudinal centerline axis **112** but do not extend circumferentially in the circumferential direction around the longitudinal centerline axis **112**. The particle separation system **1090** and the heat exchanger **1092** are provided within the outer casing **1018** of the turbo-engine **1016** and can be located at any position within the circumferential direction.

[0088] The particle separation system **1090** is disposed upstream of the heat exchanger **1092**. The particle separation system **1090** and the heat exchanger **1092** are positioned within a passage **1019** of the outer casing **1018** of the turbo-engine **1016** such that incoming airflow **1099** inside of the nacelle **1050** (which corresponds to a portion of first portion of air **1062**) enters the passage **1019** of the outer casing **1018** of the turbo-engine **1016** to first encounter the particle separation system **1090**. The particle separation system **1090** is configured to separate particles present in the incoming airflow **1091** to provide cleaner airflow (airflow having substantially less particles). The cleaner airflow then continues towards the heat exchanger **1092** to cool down a fluid (not shown in in FIG. **10**) circulating within the heat exchanger **1092** before exiting as output airflow **1100** to inside of the nacelle **1050**. The incoming airflow **1099** can be part of the first portion of air **1062**. The output airflow **1100** can also be part of the first portion of air **1062**.

[0089] As shown in FIGS. **10**, **11**, and **12**, the particle separation system **1090** and the heat exchanger **1092** can be placed at various locations with the gas turbine engine **1000**, **1001**, **1003**. The particle separation system **1090** and the heat exchanger **1092** can also be provided at any one or more of the locations shown in FIGS. **10**, **11** and **12**. For example, the particle separation system **1090** together with the heat exchanger **1092** can be provided at each of the locations shown in FIGS. **10**, **11** and **12**.

[0090] Further aspects are provided by the subject matter of the following clauses.

[0091] A particle separation system includes at least one inertial particle separator spaced apart from a heat exchanger, or integrated as a unitary piece with the heat exchanger, the at least one inertial particle separator being disposed upstream of the heat exchanger. The one or more inertial particle separator is configured and arranged to direct a first fluid flow containing a first amount of particles to an inner passage provided at a hub of the heat exchanger, an outer passage provided at a tip of the heat exchanger, or both, the hub and the tip being located at a radial periphery of the heat exchanger, and to direct a second fluid flow containing substantially no particles or a second amount of particles to a central portion of the heat exchanger to cool down a fluid circulating within the heat exchanger, the second amount of particles being substantially less than the first amount of particles.

[0092] The particle separation system of the preceding clause 2, wherein the first amount of particles in the first fluid flow has a concentration of particles greater than about 80% an original concentration of particles in an incoming fluid flow incident on the particle separation system and the second fluid flow has a concentration of particles less than about 20% the original concentration of particles in the incoming fluid flow incident on the particle separation system.

[0093] The particle separation system of any preceding clause, wherein the at least one inertial particle separator includes one or more masses, the one or more masses being disposed upstream of the heat exchanger, the one or more masses having a shape selected to deflect the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both.

[0094] The particle separation system of any preceding clause, wherein the shape is triangular, pointed, oval, or rounded.

[0095] The particle separation system of any preceding clause, wherein the at least one inertial particle separator includes one or more louvers spaced apart from the heat exchanger or integrated as a unitary piece with the heat exchanger, the one or more louvers being disposed upstream of the heat exchanger. The one or more louvers are configured and arranged to direct the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both, and to direct the second fluid flow containing substantially no particles or the second amount of particles to the central portion of the heat exchanger to cool down the fluid circulating within the heat exchanger.

[0096] The particle separation system of the preceding clause, further including an actuator configured to actuate the one or more louvers. The one or more louvers are mounted to the heat exchanger and are rotatable using the actuator.

[0097] The particle separation system of any preceding clause, wherein the at least one louver includes a first louver oriented at a first angle relative to an axial direction of the heat exchanger and a second louver oriented at a second angle relative to the axial direction of the heat exchanger, the first angle being different from the second angle, wherein the first angle and the second angle are adjustable to enhance efficiency of separating particles.

[0098] The particle separation system of the preceding clause, wherein the first louver and the second louver are spaced apart from each other and a distance separating the first louver and the second louver is adjustable to enhance an efficiency of separating particles.

[0099] The particle separation system of any preceding clause, wherein the at least one louver includes one or more passages configured to circulate the fluid circulating within the heat exchanger to cool down the fluid.

[0100] The particle separation system of any preceding clause, wherein the at least one louver having the one or more passages is further configured to melt ice particles in an incoming fluid flow incident on the particle separation system and to limit exposure of the heat exchanger to the ice particles.

[0101] The particle separation system of any preceding clause, further including a housing defining an circular interior, wherein the at least one inertial particle separator includes a vortex generator disposed within the housing, the vortex generator including a spiral coil coupled to a shaft that is configured to rotate the spiral coil, wherein the vortex generator is configured to move particles in an incoming airflow incident on the particle separator system towards a radial outer periphery of the housing by a centrifugal force generated by a rotation of the vortex generator to provide a first airflow containing a first amount of particles and to allow a second airflow having a second amount of particles to pass through the vortex generator to continue towards a heat exchanger to cool down a fluid circulating within the heat exchanger.

[0102] A particle separation system includes a housing defining an circular interior, and a vortex generator disposed within the housing. The vortex generator includes a spiral coil coupled to a shaft that is configured to rotate the spiral coil. The vortex generator is configured to move particles in an incoming fluid flow incident on the particle separation system towards a radial outer periphery of the housing by a centrifugal force generated by a rotation of the vortex generator to provide a first fluid flow containing a first amount of particles and to allow a second fluid flow having a second amount of particles to pass through the vortex generator to continue towards a heat exchanger to cool down a fluid circulating within the heat exchanger.

[0103] The particle separation system of the preceding clause, wherein the first amount of particles in the first fluid flow has a concentration of particles greater than about 80% an original concentration of particles in an incoming fluid flow incident on the particle separation system and the second fluid flow has a concentration of particles less than about 20% the original concentration of particles in the incoming fluid flow incident on the particle separation system.

[0104] A gas turbine engine includes a heat exchanger and a particle separation system including at least one inertial particle separator spaced apart from the heat exchanger or integrated as a unitary piece with the heat exchanger, the at least one inertial particle separator being disposed upstream of the heat exchanger. The at least one inertial particle separator is configured and arranged to direct a first fluid flow containing a first amount of particles to an inner passage provided at a hub of the heat exchanger, an outer passage provided at a tip of the heat exchanger, or both, the hub and the tip being located at a radial periphery of the heat exchanger, and to direct a second fluid flow containing substantially no particles or a second amount of particles to a central portion of the heat exchanger to cool down a fluid circulating within the heat exchanger, the second amount of particles being substantially less than the first amount of particles.

[0105] The gas turbine engine of the preceding clause, wherein the first amount of particles in the first fluid flow has a concentration of particles greater than about 80% an original concentration of particles in an incoming fluid flow incident on the particle separation system and the second fluid flow has a concentration of particles less than about 20% the original concentration of particles in the incoming fluid flow incident on the particle separation system.

[0106] The gas turbine engine of any preceding clause, wherein the at least one inertial particle separator includes one or more masses, the one or more masses being disposed upstream of the heat exchanger, the one or more masses having a shape selected to deflect the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both.

[0107] The gas turbine engine of any preceding clause, wherein the at least one inertial particle separator includes one or more louvers spaced apart from the heat exchanger or integrated as a unitary piece with the heat exchanger, the one or more louvers being disposed upstream of the heat exchanger. The one or more louvers are configured and arranged to direct the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both, and to direct the second fluid flow containing substantially no particles or the second amount of particles to the central portion of the heat exchanger to cool down the fluid circulating within the heat exchanger.

[0108] The gas turbine engine of the preceding clause, wherein the at least one louver includes a first louver oriented at a first angle relative to an axial direction of the heat exchanger and a second louver oriented at a second angle relative to the axial direction of the heat exchanger, the first angle being different from the second angle, wherein the first angle and the second angle are adjustable to enhance efficiency of separating particles.

[0109] The gas turbine engine of any preceding clause, wherein the first louver and the second louver are spaced apart from each other and a distance separating the first louver and the second louver is adjustable to enhance an efficiency of separating particles.

[0110] The gas turbine engine of any preceding clause, wherein the at least one louver includes one or more passages configured to circulate the fluid circulating within the heat exchanger to cool down the fluid.

[0111] The gas turbine engine of any preceding clause, wherein the at least one louver having the one or more passages is further configured to melt ice particles in an incoming fluid flow incident on the particle separation system and limit exposure of the heat exchanger to the ice particles.

[0112] The gas turbine engine of any preceding clause, wherein the heat exchanger and the particle separation system are provided within a nacelle of the gas turbine engine.

[0113] The gas turbine engine of any preceding clause, wherein the particle separation system is positioned upstream of the heat exchanger within a passage of the nacelle such that incoming fluid flow outside of the nacelle enters the passage within the nacelle to first encounter the particle separation system to provide cleaner fluid flow that continues towards the heat exchanger to cool down a fluid circulating within the heat exchanger before exiting as output fluid flow to outside of the nacelle.

[0114] The gas turbine engine of any preceding clause, wherein the particle separation system is positioned upstream of the heat exchanger within a passage of the nacelle such that incoming fluid flow inside of the nacelle enters the passage within the nacelle to first encounter the particle separation system to provide cleaner fluid flow that continues towards the heat exchanger to cool down a fluid circulating within the heat exchanger before exiting as output fluid flow to inside of the nacelle.

[0115] The gas turbine engine of any preceding clause, wherein the particle separation system is positioned upstream of the heat exchanger inside a passage within an outer casing of a turbo-engine of the gas turbine engine such that incoming fluid flow inside of the nacelle enters the passage of the outer casing of the turbo-engine to first encounter the particle separation system to provide cleaner fluid flow that continues towards the heat exchanger to cool down a fluid circulating within the heat exchanger before exiting as output fluid flow to inside of the nacelle.
[0116] A method of separating particles in a fluid flow, the method includes disposing a particle separator upstream of a heat exchanger, directing with the particle separator a first fluid flow containing a first amount of particles to an inner passage provided at a hub of the heat exchanger, an outer passage provided at a tip of the heat exchanger, or both, the hub and the tip being located at a radial periphery of the heat exchanger, and directing a second fluid flow containing substantially no particles or a second amount of particles to a central portion of the heat exchanger to cool down a fluid circulating within the heat exchanger, the second amount of particles being substantially less than the first amount of particles.
[0117] A method of separating particles in a fluid flow, the method includes disposing a vortex generator within a housing defining an circular interior, the vortex generator comprising a spiral coil coupled to a shaft that is configured to rotate the spiral coil, rotating the spiral coil, moving particles in an incoming fluid flow towards a radial outer periphery of the housing by a centrifugal force generated by the rotation of the spiral coil to provide a first fluid flow containing a first amount of particles, and passing a second fluid flow having a second amount of particles through the vortex generator to continue towards a heat exchanger to cool down a fluid circulating within the heat exchanger.
[0118] The term “fluid” is used herein broadly to include air or any other gas or a liquid, or a combination of a gas (e.g., air) and a liquid.
[0119] Although the foregoing description is directed to the preferred embodiments, other variations and modifications will be apparent to those skilled in the art, and may be made without departing from the disclosure. Moreover, features described in connection with one embodiment may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A particle separation system comprising: at least one inertial particle separator spaced apart from a heat exchanger or integrated as a unitary piece with the heat exchanger, the at least one inertial particle separator being disposed upstream of the heat exchanger, wherein the at least one inertial particle separator that is configured and arranged: to direct a first fluid flow containing a first amount of particles to an inner passage provided at a hub of the heat exchanger, an outer passage provided at a tip of the heat exchanger, or both, the hub and the tip being located at a radial periphery of the heat exchanger, and to direct a second fluid flow containing substantially no particles or a second amount of particles to a central portion of the heat exchanger to cool down a fluid circulating within the heat exchanger, the second amount of particles being substantially less than the first amount of particles.
2. The particle separation system according to claim 1, wherein the first amount of particles in the first fluid flow has a concentration of particles greater than about 80% an original concentration of particles in an incoming fluid flow incident on the particle separation system and the second fluid flow has a concentration of particles less than about 20% the original concentration of particles in the incoming fluid flow incident on the particle separation system.
3. The particle separation system according to claim 1, wherein the at least one inertial particle separator comprises one or more masses, the one or more masses being disposed upstream of the heat exchanger, the one or more masses having a shape selected to deflect the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat

exchanger, the outer passage provided at the tip of the heat exchanger, or both.
4. The particle separation system according to claim 3, wherein the shape is triangular, pointed, oval, or rounded.
5. The particle separation system according to claim 1, wherein the at least one inertial particle separator comprises one or more louvers spaced apart from the heat exchanger or integrated as a unitary piece with the heat exchanger, the one or more louvers being disposed upstream of the heat exchanger, wherein the one or more louvers are configured and arranged: to direct the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both, and to direct the second fluid flow containing substantially no particles or the second amount of particles to the central portion of the heat exchanger to cool down the fluid circulating within the heat exchanger.
6. The particle separation system according to claim 5, further comprising an actuator configured to actuate the one or more louvers, wherein the one or more louvers are mounted to the heat exchanger and are rotatable using the actuator.
7. The particle separation system according to claim 5, wherein the one or more louvers comprises a first louver oriented at a first angle relative to an axial direction of the heat exchanger and a second louver oriented at a second angle relative to the axial direction of the heat exchanger, the first angle being different from the second angle, wherein the first angle and the second angle are adjustable to enhance an efficiency of separating particles.
8. The particle separation system according to claim 7, wherein the first louver and the second louver are spaced apart from each other and a distance separating the first louver and the second louver is adjustable to enhance the efficiency of separating particles.
9. The particle separation system according to claim 5, wherein the one or more louvers comprises one or more passages configured to circulate the fluid circulating within the heat exchanger to cool down the fluid.
10. The particle separation system according to claim 9, wherein the one or more louvers having the one or more passages is further configured to melt ice particles in an incoming fluid flow incident on the particle separation system and limit exposure of the heat exchanger to the ice particles.
11. A particle separation system comprising: a housing defining an circular interior; and a vortex generator disposed within the housing, the vortex generator comprising a spiral coil coupled to a shaft that is configured to rotate the spiral coil, wherein the vortex generator is configured to move particles in an incoming fluid flow incident on the particle separation system towards a radial outer periphery of the housing by a centrifugal force generated by a rotation of the vortex generator to provide a first fluid flow containing a first amount of particles and to allow a second fluid flow having a second amount of particles to pass through the vortex generator to continue towards a heat exchanger to cool down a fluid circulating within the heat exchanger.
12. The particle system according to claim 11, wherein the first amount of particles in the first fluid flow has a concentration of particles greater than about 80% an original concentration of particles in the incoming fluid flow incident on the particle separation system and the second fluid flow has a concentration of particles less than about 20% the original concentration of particles in the incoming fluid flow incident on the particle separation system.
13. A gas turbine engine comprising: a heat exchanger; and a particle separation system comprising at least one inertial particle separator spaced apart from the heat exchanger or integrated as a unitary piece with the heat exchanger, the at least one inertial particle separator being disposed upstream of the heat exchanger, wherein the at least one inertial particle separator is configured and arranged: to direct a first fluid flow containing a first amount of particles to an inner passage provided at a hub of the heat exchanger, an outer passage provided at a tip of the heat exchanger, or both, the hub and the tip being located at a radial periphery of the heat exchanger; and to direct a second fluid flow containing substantially no particles or a second amount of particles to a central

portion of the heat exchanger to cool down a fluid circulating within the heat exchanger, the second amount of particles being substantially less than the first amount of particles.

14. The gas turbine engine according to claim 13, wherein the first amount of particles in the first fluid flow has a concentration of particles greater than about 80% an original concentration of particles in an incoming fluid flow incident on the particle separation system and the second fluid flow has a concentration of particles less than about 20% the original concentration of particles in the incoming fluid flow incident on the particle separation system.

15. The gas turbine engine according to claim 13, wherein the at least one inertial particle separator comprises one or more masses, the one or more masses being disposed upstream of the heat exchanger, the one or more masses having a shape selected to deflect the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both.

16. The gas turbine engine according to claim 13, wherein the at least one inertial particle separator comprises one or more louvers spaced apart from the heat exchanger or integrated as a unitary piece with the heat exchanger, the one or more louvers being disposed upstream of the heat exchanger, wherein the one or more louvers are configured and arranged: to direct the first fluid flow containing the first amount of particles to the inner passage provided at the hub of the heat exchanger, the outer passage provided at the tip of the heat exchanger, or both; and to direct the second fluid flow containing substantially no particles or the second amount of particles to the central portion of the heat exchanger to cool down the fluid circulating within the heat exchanger.

17. The gas turbine engine according to claim 16, wherein the one or more louvers comprises a first louver oriented at a first angle relative to an axial direction of the heat exchanger and a second louver oriented at a second angle relative to the axial direction of the heat exchanger, the first angle being different from the second angle, wherein the first angle and the second angle are adjustable to enhance an efficiency of separating particles.

18. The gas turbine engine according to claim 17, wherein the first louver and the second louver are spaced apart from each other and a distance separating the first louver and the second louver is adjustable to enhance the efficiency of separating particles.

19. The gas turbine engine according to claim 16, wherein the one or more louvers comprises one or more passages configured to circulate the fluid circulating within the heat exchanger to cool down the fluid.

20. The gas turbine engine according to claim 19, wherein the one or more louvers having the one or more passages is further configured to melt ice particles in an incoming fluid flow incident on the particle separation system and limit exposure of the heat exchanger to the ice particles.